

Vivek Rathia

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Education

Amity University, Noida <i>M.Tech in Artificial Intelligence</i>	2025 – 2027 CGPA: 7.32 / 10.00
Bhilai Institute of Technology, Durg <i>B.Tech in Computer Science and Engineering</i>	2020 – 2024 CGPA: 8.33 / 10.00

Projects

Mentor Management System | Svelte.js, Node.js, TypeScript, MongoDB, Tailwind CSS

- Designed and developed a **full-stack mentorship management platform** with a **Svelte.js** frontend and **RESTful APIs** using **Node.js** and **TypeScript**.
- Structured the application around **reusable components** and **reactive state management** in Svelte for seamless user interaction.
- Implemented **secure authentication** and **role-based access control** using **Firebase Auth** and **JWT**.
- Integrated **MongoDB** to store mentor portfolios, session logs, and student progress metrics, enabling efficient data retrieval and scalability.

Currency Converter Using React | React, Tailwind CSS, Node.js, REST API

- Engineered a **real-time currency converter** with **modular React components** and **Tailwind CSS** for a sleek and responsive UI.
- Consumed **third-party currency exchange APIs** via **Axios**, handling asynchronous data and error states efficiently.
- Implemented **custom hooks** and **controlled components** for reactive form management and real-time validation.
- Designed **backend service prototypes** using **Node.js** and **Express** for potential API caching and rate-limiting support.

Movie Recommender System | Python, Flask, Pandas, scikit-learn, MongoDB

- Built a **content-based recommender system** leveraging **cosine similarity** over **TF-IDF vectors** derived from movie metadata.
- Preprocessed and normalized large datasets using **Pandas** and **scikit-learn** for feature extraction and vectorization.
- Deployed the model via a **Flask REST API** with endpoints for **personalized movie recommendations** and feedback submission.
- Stored user interactions and feedback data in **MongoDB**, enabling future improvements with collaborative filtering approaches.

Netflix Dataset Analysis | Python, Jupyter, Pandas, Seaborn, MongoDB

- Performed **in-depth exploratory data analysis (EDA)** on Netflix's global catalogue using **Pandas** and **Seaborn**.
- Cleaned and transformed unstructured data for visualization, identifying patterns in **content release**, **genre trends**, and **regional availability**.
- Imported cleaned data into **MongoDB collections** for querying with **aggregation pipelines** and building dashboards.
- Prepared analytical insights to support **data-driven decisions** on content acquisition and user targeting strategies.

Technical Skills

Languages and Frameworks: TypeScript, JavaScript, C++, Python, HTML, CSS, Svelte.js, React.js, Node.js, Express.js, Tailwind CSS, Bootstrap

Databases & Cloud: MongoDB, MySQL, Firebase, NoSQL, SQL, PostgreSQL, Google Cloud Platform (GCP), Firebase Realtime DB, Firestore

Tools & Technologies: Git, REST APIs, Postman, JWT, Docker, VS Code, Figma, GitHub, Agile, Webpack, Vite, CI/CD

Data Science & ML: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Tableau, Power BI